

Total No. of Questions : 8]

SEAT No. :

PE4256

[Total No. of Pages : 2

[6582]-27

S.E. (Electronics/E & TC) (Electronics & Computer
Engg. Electronics (VLSI Design & Tech.)/E.C(A.C.T.))

DIGITAL CIRCUITS

(2019 Pattern) (Semester - III) (204182)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

Q1) a) Design of one-bit magnitude comparator. [6]

b) Implement the following expression using single 8 : 1 multiplexer [6]

$$Y = \sum m(0,1,2,5,7,8,9,14,15)$$

c) Write a short note on Look ahead carry generator. [6]

OR

Q2) a) Design of 3 bit even parity generator. [6]

b) Implement the following function using 1 : 8 De-multiplexer. [6]

$$F1(A,B,C) = \sum m(0,3,7)$$

$$F2(A,B,C) = \sum m(1,2,5)$$

$$F3(A,B,C) = \sum m(1,3,6)$$

c) Write a short note on ALU. [6]

Q3) a) Define the following terms as applied to flip-flop. [6]

i) Set up time

ii) Hold time

iii) Propagation Delays

b) Perform the JK flip-flop to D flip-flop conversion. [6]

c) Design and implement a MOD 6 asynchronous counter IC 7490. [5]

OR

P.T.O.

- Q4)** a) Design a sequence generator to generate sequence 11001 using shift register. [6]
 b) Design and implement a MOD 4 synchronous counter using D flipflop. [6]
 c) Explain the various modes of Shift Register. [5]

- Q5)** a) Design sequence detector to detect a sequence 101... using D flip-flops. Use mealy machine. [9]
 b) Draw an ASM chart for a 2-bit binary counter having enable line E such that E=1 (counting enabled) and E = 0 (hold present count). [8]

OR

- Q6)** a) Design sequence detector to detect a sequence 110... using T flip-flops. Use mealy machine. [9]
 b) Draw an ASM chart and state table for a 2-bit up-down counter having mode control input : [8]

M = 1 : Up counting

M = 0 : Down counting

The circuit should generate a output 1 whenever count becomes minimum or maximum.

- Q7)** a) Compare between PROM, PAL, PLA [6]
 b) Design following functions using PLA. [6]

$$F_1 = \sum m(1, 3, 5) \quad F_2 = \sum m(5, 6, 7)$$

Implement the circuit with a PLA having 3 inputs, 3 product terms and 2 output.

- c) Draw and explain the architecture of CPLD. [6]

OR

- Q8)** a) Draw a neat diagram of one cell of static RAM and explain its working. [6]
 b) Design following functions using PLA : [6]

$$F_1 = \sum m(3, 5, 7) \quad F_2 = \sum m(4, 5, 7)$$

- c) Draw and explain the architecture of FPGA. [6]

x

x

x